

```
In [20]: import pandas as pd
import numpy as np
```

```
In [21]: books = pd.read_csv("C:\\Users\\sanje\\Downloads\\Book Recom\\Books.csv\\Books.csv"
users = pd.read_csv("C:\\Users\\sanje\\Downloads\\Book Recom\\Users.csv\\Users.csv"
ratings = pd.read_csv("C:\\Users\\sanje\\Downloads\\Book Recom\\Ratings.csv\\Rating
```

C:\Users\sanje\AppData\Local\Temp\ipykernel_14328\92926672.py:1: DtypeWarning: Columns (3) have mixed types. Specify dtype option on import or set low_memory=False.
books = pd.read_csv("C:\\Users\\sanje\\Downloads\\Book Recom\\Books.csv\\Books.csv")

```
In [22]: books.head()
```

```
Out[22]:
```

	ISBN	Book-Title	Book-Author	Year-Of-Publication	Publisher	
0	0195153448	Classical Mythology	Mark P. O. Morford	2002	Oxford University Press	http://images.amazon.com/images/P
1	0002005018	Clara Callan	Richard Bruce Wright	2001	HarperFlamingo Canada	http://images.amazon.com/images/P
2	0060973129	Decision in Normandy	Carlo D'Este	1991	HarperPerennial	http://images.amazon.com/images/P
3	0374157065	Flu: The Story of the Great Influenza Pandemic...	Gina Bari Kolata	1999	Farrar Straus Giroux	http://images.amazon.com/images/P
4	0393045218	The Mummies of Urumchi	E. J. W. Barber	1999	W. W. Norton & Company	http://images.amazon.com/images/P

```
In [23]: users.head()
```

```
Out[23]:
```

	User-ID	Location	Age
0	1	nyc, new york, usa	NaN
1	2	stockton, california, usa	18.0
2	3	moscow, yukon territory, russia	NaN
3	4	porto, v.n.gaia, portugal	17.0
4	5	farnborough, hants, united kingdom	NaN

```
In [24]: ratings.head()
```

```
Out[24]:
```

	User-ID	ISBN	Book-Rating
0	276725	034545104X	0
1	276726	0155061224	5
2	276727	0446520802	0
3	276729	052165615X	3
4	276729	0521795028	6

```
In [25]: print(books.shape)
print(ratings.shape)
print(users.shape)
```

```
(271360, 8)
(1149780, 3)
(278858, 3)
```

```
In [26]: books.isnull().sum()
```

```
Out[26]: ISBN                0
Book-Title                0
Book-Author              1
Year-Of-Publication      0
Publisher                 2
Image-URL-S              0
Image-URL-M              0
Image-URL-L              3
dtype: int64
```

```
In [27]: users.isnull().sum()
```

```
Out[27]: User-ID          0
Location          0
Age             110762
dtype: int64
```

```
In [28]: ratings.isnull().sum()
```

```
Out[28]: User-ID          0
ISBN            0
Book-Rating     0
dtype: int64
```

```
In [29]: books.duplicated().sum()
```

```
Out[29]: 0
```

```
In [30]: ratings.duplicated().sum()
```

```
Out[30]: 0
```

```
In [31]: users.duplicated().sum()
```

```
Out[31]: 0
```

```
In [32]: books.columns
```

```
Out[32]: Index(['ISBN', 'Book-Title', 'Book-Author', 'Year-Of-Publication', 'Publisher',
               'Image-URL-S', 'Image-URL-M', 'Image-URL-L'],
              dtype='object')
```

```
In [33]: books.dtypes
```

```
Out[33]: ISBN                object
Book-Title                object
Book-Author              object
Year-Of-Publication      object
Publisher                 object
Image-URL-S              object
Image-URL-M              object
Image-URL-L              object
dtype: object
```

```
In [34]: users.columns
```

```
Out[34]: Index(['User-ID', 'Location', 'Age'], dtype='object')
```

```
In [35]: users.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 278858 entries, 0 to 278857
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   User-ID     278858 non-null  int64
1   Location    278858 non-null  object
2   Age         168096 non-null  float64
dtypes: float64(1), int64(1), object(1)
memory usage: 6.4+ MB
```

```
In [36]: users.isnull().sum()
```

```
Out[36]: User-ID          0
Location          0
Age             110762
dtype: int64
```

```
In [37]: users['Age'].describe()
```

```
Out[37]: count      168096.000000
mean           34.751434
std            14.428097
min             0.000000
25%            24.000000
50%            32.000000
75%            44.000000
max            244.000000
Name: Age, dtype: float64
```

```
In [38]: users[(users['Age'] >= 10) & (users['Age'] < 20)].shape[0]
```

```
Out[38]: 18894
```

```
In [39]: users[(users['Age'] >= 30) & (users['Age'] < 40)].shape[0]
```

```
Out[39]: 41889
```

```
In [40]: users['Age'].max()
```

```
Out[40]: 244.0
```

```
In [41]: bins = [0, 10, 20, 30, 40, 50, 200]
labels = ['0-10', '10-20', '20-30', '30-40', '40-50', '50+']
users['Age_Range'] = pd.cut(users['Age'], bins=bins, labels=labels)
```

```
In [42]: users['Age_Range'].value_counts().sort_index()
```

```
Out[42]: 0-10      737
          10-20    22866
          20-30    52261
          30-40   40253
          40-50   25898
          50+     25639
          Name: Age_Range, dtype: int64
```

```
In [43]: ratings_with_name = ratings.merge(books,on ="ISBN")
```

```
In [44]: num_rating_df = ratings_with_name.groupby('Book-Title').count()['Book-Rating'].reset_index()
          num_rating_df.rename(columns ={'Book-Rating': 'num_rating'},inplace =True)
```

```
In [45]: num_rating_df
```

```
Out[45]:
```

	Book-Title	num_rating
0	A Light in the Storm: The Civil War Diary of ...	4
1	Always Have Popsicles	1
2	Apple Magic (The Collector's series)	1
3	Ask Lily (Young Women of Faith: Lily Series, ...	1
4	Beyond IBM: Leadership Marketing and Finance ...	1
...	...	...
241066	Ä?Ä?piraten.	2
241067	Ä?Ä?rger mit Produkt X. Roman.	4
241068	Ä?Ä?sterlich leben.	1
241069	Ä?Ä?stlich der Berge.	3
241070	Ä?Ä?thique en toc	2

241071 rows × 2 columns

```
In [46]: avg_rating_df = ratings_with_name.groupby('Book-Title').mean()['Book-Rating'].reset_index()
          avg_rating_df.rename(columns ={'Book-Rating': 'avg_rating'},inplace =True)
```

```
C:\Users\sanje\AppData\Local\Temp\ipykernel_14328\3262757628.py:1: FutureWarning:
The default value of numeric_only in DataFrameGroupBy.mean is deprecated. In a future version, numeric_only will default to False. Either specify numeric_only or select only columns which should be valid for the function.
  avg_rating_df = ratings_with_name.groupby('Book-Title').mean()['Book-Rating'].reset_index()
```

Out[46]:

	Book-Title	avg_rating
0	A Light in the Storm: The Civil War Diary of ...	2.250000
1	Always Have Popsicles	0.000000
2	Apple Magic (The Collector's series)	0.000000
3	Ask Lily (Young Women of Faith: Lily Series, ...	8.000000
4	Beyond IBM: Leadership Marketing and Finance ...	0.000000
...	...	...
241066	Ä?Ä?piraten.	0.000000
241067	Ä?Ä?rger mit Produkt X. Roman.	5.250000
241068	Ä?Ä?sterlich leben.	7.000000
241069	Ä?Ä?stlich der Berge.	2.666667
241070	Ä?Ä?thique en toc	4.000000

241071 rows × 2 columns

```
In [47]: popular_df = num_rating_df.merge(avg_rating_df,on = 'Book-Title')
popular_df
```

Out[47]:

	Book-Title	num_rating	avg_rating
0	A Light in the Storm: The Civil War Diary of ...	4	2.250000
1	Always Have Popsicles	1	0.000000
2	Apple Magic (The Collector's series)	1	0.000000
3	Ask Lily (Young Women of Faith: Lily Series, ...	1	8.000000
4	Beyond IBM: Leadership Marketing and Finance ...	1	0.000000
...	...	...	...
241066	Ä?Ä?piraten.	2	0.000000
241067	Ä?Ä?rger mit Produkt X. Roman.	4	5.250000
241068	Ä?Ä?sterlich leben.	1	7.000000
241069	Ä?Ä?stlich der Berge.	3	2.666667
241070	Ä?Ä?thique en toc	2	4.000000

241071 rows × 3 columns

```
In [48]: popular_df = popular_df[popular_df['num_rating']>=250].sort_values('avg_rating',asc
```

```
In [49]: popular_df.merge(books,on='Book-Title').drop_duplicates('Book-Title')[['Book-Title'
```

Out[49]:

	Book-Title	Book-Author	Image-URL-M	num_rating	avg
0	Harry Potter and the Prisoner of Azkaban (Book 3)	J. K. Rowling	http://images.amazon.com/images/P/0439136350.0...	428	5.
3	Harry Potter and the Goblet of Fire (Book 4)	J. K. Rowling	http://images.amazon.com/images/P/0439139597.0...	387	5.
5	Harry Potter and the Sorcerer's Stone (Book 1)	J. K. Rowling	http://images.amazon.com/images/P/0590353403.0...	278	5.
9	Harry Potter and the Order of the Phoenix (Boo...	J. K. Rowling	http://images.amazon.com/images/P/043935806X.0...	347	5.
13	Harry Potter and the Chamber of Secrets (Book 2)	J. K. Rowling	http://images.amazon.com/images/P/0439064872.0...	556	5.
16	The Hobbit : The Enchanting Prelude to The Lor...	J.R.R. TOLKIEN	http://images.amazon.com/images/P/0345339681.0...	281	5.
17	The Fellowship of the Ring (The Lord of the Ri...	J.R.R. TOLKIEN	http://images.amazon.com/images/P/0345339703.0...	368	4.
26	Harry Potter and the Sorcerer's Stone (Harry P...	J. K. Rowling	http://images.amazon.com/images/P/059035342X.0...	575	4.
28	The Two Towers (The Lord of the Rings, Part 2)	J.R.R. TOLKIEN	http://images.amazon.com/images/P/0345339711.0...	260	4.
39	To Kill a Mockingbird	Harper Lee	http://images.amazon.com/images/P/0446310786.0...	510	4.
47	The Da Vinci Code	Dan Brown	http://images.amazon.com/images/P/0385504209.0...	898	4.
53	The Five People You Meet in Heaven	Mitch Albom	http://images.amazon.com/images/P/0786868716.0...	430	4.

	Book-Title	Book-Author	Image-URL-M	num_rating	avg
55	The Catcher in the Rye	J.D. Salinger	http://images.amazon.com/images/P/0316769487.0...	449	4.
62	The Lovely Bones: A Novel	Alice Sebold	http://images.amazon.com/images/P/0316666343.0...	1295	4.
63	1984	George Orwell	http://images.amazon.com/images/P/0451524934.0...	284	4.
72	Prodigal Summer: A Novel	Barbara Kingsolver	http://images.amazon.com/images/P/0060959037.0...	253	4.
73	Neverwhere	Neil Gaiman	http://images.amazon.com/images/P/0380789019.0...	265	4.
78	The Secret Life of Bees	Sue Monk Kidd	http://images.amazon.com/images/P/0142001740.0...	774	4.
84	Stupid White Men ...and Other Sorry Excuses fo...	Michael Moore	http://images.amazon.com/images/P/0060392452.0...	283	4.
85	Tuesdays with Morrie: An Old Man, a Young Man,...	MITCH ALBOM	http://images.amazon.com/images/P/0385484518.0...	493	4.
88	The Red Tent (Bestselling Backlist)	Anita Diamant	http://images.amazon.com/images/P/0312195516.0...	723	4.
89	The Hitchhiker's Guide to the Galaxy	Douglas Adams	http://images.amazon.com/images/P/0671461494.0...	268	4.
98	Nickel and Dimed: On (Not) Getting By in America	Barbara Ehrenreich	http://images.amazon.com/images/P/0805063897.0...	335	4.
100	Into the Wild	Jon Krakauer	http://images.amazon.com/images/P/0385486804.0...	252	4.
103	Fahrenheit 451	Ray Bradbury	http://images.amazon.com/images/P/3257208626.0...	409	4.
116	Girl with a Pearl Earring	Tracy Chevalier	http://images.amazon.com/images/P/0452282152.0...	526	4.
117	Outlander	DIANA GABALDON	http://images.amazon.com/images/P/0440222915.0...	283	4.
123	Where the Heart Is (Oprah's Book Club (Paperba...	Billie Letts	http://images.amazon.com/images/P/0446672211.0...	585	4.
124	Seabiscuit: An	LAURA HILLENBRAND	http://images.amazon.com/images/P/0449005615.0...	275	4.

	Book-Title	Book-Author	Image-URL-M	num_rating	avg
	American Legend				
126	Life of Pi	Yann Martel	http://images.amazon.com/images/P/0151008116.0...	664	4.
130	The Bean Trees	Barbara Kingsolver	http://images.amazon.com/images/P/0060915544.0...	389	4.
133	A Child Called \It\": One Child's Courage to S...	Dave Pelzer	http://images.amazon.com/images/P/1558743669.0...	265	4.
135	ANGELA'S ASHES	Frank McCourt	http://images.amazon.com/images/P/0684874350.0...	279	4.
137	Good in Bed	Jennifer Weiner	http://images.amazon.com/images/P/0743418174.0...	490	4.
139	Fast Food Nation: The Dark Side of the All-Ame...	Eric Schlosser	http://images.amazon.com/images/P/0060938455.0...	321	4.
140	American Gods	Neil Gaiman	http://images.amazon.com/images/P/0380789035.0...	302	4.
141	Skippping Christmas	JOHN GRISHAM	http://images.amazon.com/images/P/0385505833.0...	322	4.
147	Left Behind: A Novel of the Earth's Last Days ...	Tim Lahaye	http://images.amazon.com/images/P/0842329129.0...	318	4.
149	The Golden Compass (His Dark Materials, Book 1)	PHILIP PULLMAN	http://images.amazon.com/images/P/037582345X.0...	336	4.
157	The Color Purple	Alice Walker	http://images.amazon.com/images/P/0671617028.0...	314	3.
163	Silence of the Lambs	Thomas Harris	http://images.amazon.com/images/P/0312924585.0...	256	3.
166	About a Boy	Nick Hornby	http://images.amazon.com/images/P/1573227331.0...	262	3.
173	Seven Up (A Stephanie Plum Novel)	Janet Evanovich	http://images.amazon.com/images/P/0312265840.0...	278	3.
175	The Alchemist: A Fable About Following Your Dream	Paulo Coelho	http://images.amazon.com/images/P/0062502174.0...	266	3.
177	Hard Eight : A Stephanie Plum Novel (A Stephan...	Janet Evanovich	http://images.amazon.com/images/P/0312983867.0...	269	3.

	Book-Title	Book-Author	Image-URL-M	num_rating	avg
182	Suzanne's Diary for Nicholas	James Patterson	http://images.amazon.com/images/P/0316969443.0...	457	3.
186	Lord of the Flies	William Gerald Golding	http://images.amazon.com/images/P/0399501487.0...	259	3.
192	A Prayer for Owen Meany	John Irving	http://images.amazon.com/images/P/0345361792.0...	413	3.
194	The Vampire Lestat (Vampire Chronicles, Book II)	ANNE RICE	http://images.amazon.com/images/P/0345313860.0...	301	3.
195	White Oleander : A Novel (Oprah's Book Club)	Janet Fitch	http://images.amazon.com/images/P/0316284955.0...	356	3.

In [50]: *#collaborative filtering*

```
In [51]: x = ratings_with_name.groupby('User-ID').count()['Book-Rating']>200
padhe_likhe_users = x[x].index
```

```
In [52]: filtered_rating = ratings_with_name[ratings_with_name['User-ID'].isin(padhe_likhe_u
```

```
In [53]: y = filtered_rating.groupby('Book-Title').count()['Book-Rating']>=50
famous_books = y[y].index
```

```
In [54]: final_ratings = filtered_rating[filtered_rating['Book-Title'].isin(famous_books)]
```

```
In [55]: pt=final_ratings.pivot_table(index= 'Book-Title', columns='User-ID', values = 'Book-R
```

```
In [56]: pt
```

Out[56]:

User-ID	254	2276	2766	2977	3363	4017	4385	6251	6323	6543	...	271705	27397
---------	-----	------	------	------	------	------	------	------	------	------	-----	--------	-------

Book-Title													
1984	9.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	10.0	NaN
1st to Die: A Novel	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	9.0	...	NaN	NaN
2nd Chance	NaN	10.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	...	NaN	NaN
4 Blondes	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	NaN	...	NaN	NaN
A Bend in the Road	0.0	NaN	7.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	0.
...	...	...	...	...	...	...	...	...	...	...	...	...	...
Year of Wonders	NaN	NaN	NaN	7.0	NaN	NaN	NaN	NaN	NaN	0.0	...	NaN	9.
You Belong To Me	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	...	NaN	NaN
Zen and the Art of Motorcycle Maintenance: An Inquiry into Values	NaN	NaN	NaN	NaN	0.0	NaN	NaN	0.0	NaN	NaN	...	NaN	NaN
Zoya	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	0.
\O\" Is for Outlaw"	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN

706 rows × 810 columns



```
In [57]: pt.fillna(0,inplace=True)
```

```
In [58]: pt
```

Out[58]:

User-ID	254	2276	2766	2977	3363	4017	4385	6251	6323	6543	...	271705	273979
---------	-----	------	------	------	------	------	------	------	------	------	-----	--------	--------

Book-Title													
1984	9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	10.0	0.0
1st to Die: A Novel	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.0	...	0.0	0.0
2nd Chance	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
4 Blondes	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
A Bend in the Road	0.0	0.0	7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
Year of Wonders	0.0	0.0	0.0	7.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	9.0
You Belong To Me	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
Zen and the Art of Motorcycle Maintenance: An Inquiry into Values	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
Zoya	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
"Is for Outlaw"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0

706 rows × 810 columns



```
In [59]: from sklearn.metrics.pairwise import cosine_similarity
```

```
In [60]: similarity_scores = cosine_similarity(pt)
```

```
In [65]: similarity_scores.shape
```

```
Out[65]: (706, 706)
```

```
In [100... def recommend(book_name):
    index = np.where(pt.index == book_name)[0][0]

    similar_items = sorted(
        list(enumerate(similarity_scores[index])),
        key=lambda x: x[1],
        reverse=True
    )[1:6]

    for i in similar_items:
        print(pt.index[i[0]])
```

```
In [101... pt.index[545]
```

```
Out[101]: "The Handmaid's Tale"
```

In [103...

```
recommend('The Da Vinci Code')
```

```
Angels & Demons  
Touching Evil  
Saving Faith  
The Sweet Potato Queens' Book of Love  
Middlesex: A Novel
```

In []: